AI-Based Electricity Demand Intelligence System For Sri Lanka

Project ID

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The production and use of electricity in Sri Lanka is a rapidly developing area as the use of renewable sources of energy and electric automobiles grows. Although such a shift and the any changes in the operations of the power system contribute to the national interests in energy security and environmental sustainability, it also leads to higher variability in the operations of the power system. Solar, Hydro and wind are the sources of renewable energy, which are intermittent in nature and result in the variations of power supply, causing difficulties in stabilizing the grid. Therefore, the proper forecasting of electricity demand has gained a higher role in long-term energy planning and real time grid operations. Nonlinear, complex and dynamic nature of electricity consumption patterns in the present is however hard to capture using traditional forecasting techniques. There is a large research gap concerning electricity prediction models that are currently applied in Sri Lanka. Most of the methods in use presuppose primarily the historical data about electricity loads and do not take into account such significant external factors as weather conditions, calendar effects, and socioeconomic changes that can impact electricity demand. Moreover, the usage of modern deep learning methods, such as hybrid CNN-LSTM or Transformer, is not much. Moreover, there are minimal probabilistic forecasting techniques, which measure the uncertainty in the demand forecasts. To overcome these shortcomings, an AI-Based Electricity Demand Intelligence System would be proposed in this study and be based on four subsections, namely Multi-Source Data Integration Engine, Hybrid Deep Learning Framework, Behavioural Analysis Module, and Probabilistic Uncertainty Quantifier. The system, implemented in Python with TensorFlow and PyTorch, combines CNN-LSTM and Transformer models as part of an ensemble of adaptive models with a Bayesian probabilistic layer with Monte Carlo Dropout to provide prediction intervals.

Keywords: Electricity Demand Forecasting, Hybrid CNN–LSTM Model, Transformer Models,

Sri Lanka Energy Sector, Probabilistic Forecasting

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LIST OF ABBREVIATION

LSTM - Long Short-Term Memory

MLP - Multilayer Perceptron

GRU - Gated Recurrent Units

MAE -Mean Absolute Error

Mean Absolute Percentage Error (MAPE)

MSE -Mean Squared Error

RMSE -Root Mean Squared Error

CEB – Ceylon Electricity Board

LECO –Lanka Electricity Company

CNN- Convolutional Neural Network

ML – Machine Learning

GPU - Graphics Processing Unit

MI -Machine Learning

AI- Artificial Intelligence

SDG-Sustainable Development Goals

IT-Information Technology

ARDL Autoregressive Distributed Lag

ARIMA – Auto Regressive Integrated Moving Average

RMSE- Root Mean Square Error

PICP -Prediction Interval Coverage Probability

FR – Functional Requirements

NFR – Non-Functional Requirement

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INTRODUCTION

1.1 Background and Context

1.1.1 Describe the real-world problem

Increasing electricity demand worldwide, driven by industrial growth and the adoption of electric technologies such as electric vehicles, places pressure on power grids and increases dependence on fossil fuel-based generation [1]. In Sri Lanka, the Ceylon Electricity Board (CEB) manages a complex national grid where demand fluctuates significantly across seasons, industrial cycles, public holidays, and weather events. Despite ongoing grid modernization efforts, the country continues to experience demand–supply mismatches, resulting in load shedding, operational inefficiencies, and economic losses. The 2022 energy crisis in Sri Lanka, partly worsened by foreign exchange shortages and inadequate demand planning, led to power cuts of up to 13 hours daily highlighting the critical need for smarter, data-driven electricity demand management [2]. The stress on power grids can cause financial strain, environmental problems, and interruptions in electricity supply that affect homes, businesses, and industries.

1.1.2 Who is affected and why it is important

1. Energy Utilities & Grid Operators (CEB): Overestimating energy reserves results in unnecessary costs, whereas underestimating demand can trigger blackouts and load shedding.
2. Industrial & Commercial Consumers: Unpredictable power interruptions disrupt manufacturing and business operations, causing economic loss.
3. Residential Consumers: Scheduled and unscheduled power cuts reduce quality of life and productivity.

1.1.3 Statistical evidence (local/global)

Globally, electricity demand is projected to rise by over 25% by 2030, driven by electrification of transport, urbanization, and industrial growth [1]. In Sri Lanka, annual electricity demand grew at an average rate of approximately 5–6% per year over the past decade [3]. Seasonal variations are pronounced: peak demand during the dry season (February–April) can exceed average demand by 15–20%, placing severe strain on hydropower-dependent generation capacity. Furthermore, a 2021 study on South Asian power grids found that improving demand forecast accuracy by just 1% can reduce operating costs by up to USD 1.6 million annually for a mid-sized utility.

1.1.4 Analysis of Existing Solutions and Their Limitations

The recent studies and application of utilities in Sri Lanka have mainly shifted their focus to less advanced statistical techniques to simplistic Machine Learning (ML) and Artificial Intelligence (AI) systems. Nevertheless, such solutions are narrow in range and technical. The existing environment is typified by the following gaps:

Architectural Stagnation: Most of the studies that are specific to Sri Lanka [4][5] use Long Short-Term Memory (LSTM) networks and Multi-Layer Perceptron (MLP). Although they are better than simple regressions, they tend to be optimized for short term predictions and fail to handle long term time dependencies and complicated geometrical correlations.

Dependency on Univariate Data: The current models are mostly "data-siloed" making use of practically only the historical load data. They do not incorporate major external drivers, including weather variables, socioeconomic changes, and policy changes, which results in models which are unable to respond to abrupt changes or black swan occurrences in the energy industry.

Lack of Hybrid and Advanced DL Models: No published Sri Lankan applications of Transformers, Gated Recurrent Units (GRU), or CNN-LSTM hybrids are found. Therefore, the advantages of Attention Mechanisms (that dwell upon particular crucial time-steps) and spatial-feature extraction have not been exploited locally.

Deterministic Output Constraints: The existing solutions offer point forecasts (one predicted value) that are assessed by means of conventional measures such as MAE and RMSE. These measures do not provide any information on uncertainty or risk which is essential in the case of CEB planners who need to control the margin of error in grid stability and load shedding

1.1.5 Ideas for the proposed solution and its parts

To fill the identified gaps, the offered solution will take a step further and leave the domain of mere prediction and design a Probabilistic Decision-Support System specifically adapted to the Sri Lankan energy environment. The solution includes of four integrated components:

Multi-Source Data Integration & Interpretability Engine: This module will not use only past demand as external features such as weather conditions (temperature, humidity), calendar effects (Poya days, school holidays) and economic indicators are ingested. It will employ Interpretable ML (e.g., SHAP values or Attention scores) to enable stakeholders to understand why a particular demand spike will happen.

Hybrid Deep Learning Framework: The basic predictive scheme will be a CNN-LSTM or Transformer-based one. The CNN layers will obtain the spatial features and local patterns of the high-frequency data, and the LSTM/Transformer components will obtain long-term seasonal dependence that is much more accurate than the regular models.

Behavioral & Anomaly Analysis Module: This will be used to categories the consumption pattern between households and industries and in particular the weekday and weekend behavior. The system offers a subtle perspective of the dynamics of "base load" as compared to that of the variable load by detecting anomalies and deviations.

Probabilistic Uncertainty Quantifier: The system will also stop using point forecasts and instead use Quantile Regression or Monte Carlo Dropout to produce a prediction interval (e.g., a 95% confidence range). This will enable decision-makers to evaluate the worst-case and best-case scenarios, enabling easier risk management of power generation and distribution.

1.1.6 Relevant SDG alignment

This project directly supports the following United Nations Sustainable Development Goals:

- → SDG 7 – Affordable and Clean Energy: By optimizing demand prediction and supporting better grid planning, the system contributes to efficient, reliable, and sustainable energy delivery.
- → SDG 9 – Industry, Innovation, and Infrastructure: The use of advanced AI techniques—adaptive forecasting, explainable AI, scenario modeling—represents a significant infrastructure innovation for energy management.
- → SDG 13 – Climate Action: Improved demand forecasting reduces over-generation from fossil fuel sources, thereby cutting unnecessary carbon emissions.

1.1.7 Research cluster alignment

This project is aligned with the CoEAI – Centre of Excellence for AI research cluster at SLIIT University, under the Information Technology (IT) specialization. The work contributes to the cluster's focus on applied machine learning, intelligent decision-support systems, and data-driven infrastructure optimization. The adaptive forecasting part specifically advances research at the intersection of time-series machine learning and intelligent energy systems.

1.2 Literature Review

Electricity demand forecasting has been widely studied using both statistical and machine learning approaches, as accurate prediction is essential for effective energy management. In recent years, deep learning models have attracted significant attention to their ability to analyze complex time-series data and capture nonlinear patterns in electricity consumption. [6]

Several studies have highlighted the effectiveness of neural network-based models in demand forecasting. For example, research comparing traditional statistical forecasting techniques with deep learning models found that the LSTM model produced the lowest prediction errors when forecasting the next day's electricity demand, demonstrating its strong ability to handle sequential energy consumption data. Similarly, another study focusing on electricity load forecasting in Sri Lanka applied deep learning techniques and reported improved forecasting accuracy compared with conventional approaches.[5]

Hybrid forecasting models have been introduced to enhance prediction performance. A study that combined the Prophet forecasting model with the XGBoost algorithm, using four years of data from the Ceylon Electricity Board, showed noticeable improvements in accuracy. The hybrid model reduced the Mean Absolute Error (MAE) by 19.45 and the Mean Absolute Percentage Error (MAPE) by 1.32% when compared with the Prophet model used independently. [4] These findings indicate that combining different machine learning techniques can significantly improve forecasting outcomes.

Artificial neural networks have also been widely applied in short-term load forecasting. A three-layer backpropagation of neural Network found to perform better than several traditional statistical models, highlighting the capacity of neural networks to capture patterns in electricity demand data. [7]

Deep learning architectures such as convolutional neural networks have been used for microgrid demand forecasting. The data derived from these studies showed lower mean absolute error and root mean square error values compared with other forecasting models, which show improved electricity consumption forecasting. An advanced Autoregressive Distributed Lag (ARDL) model that includes weather-related variables to predict electricity consumption. Their findings showed that the ARDL model provided the most accurate monthly consumption forecasts when compared with seven other forecasting methods, emphasizing the influence of environmental factors on electricity demand. In conclusion, all these earlier studies prove that advanced forecasting approaches, notably deep learning and hybrid machine learning models, provide better prediction of accuracy than traditional statistical methods. These improvements highlight the importance of adaptive forecasting and prediction systems in improving electricity demand forecasting and supporting efficient energy planning.[8]

Keywords – Electricity demand forecasting, AI based smart grid management, Electric load pattern analysis, AI based energy management system, Ceylon Electricity Board electricity demand, Energy demand prediction using LSTM

1.3 Research Gap

Majority of the AI-based electricity demand forecasting research in Sri Lanka aim at enhancing the accuracy of the prediction via LSTM, ARIMA, as well as gradient boosting. Nevertheless, they do not explore the causes of demand fluctuations and effective demand management strategies. This knowledge gap can be summed up under three key gaps in the literature:

Key gaps

- **Behavioral Analysis**: Existing models do not categorize day-to-day habits of consumption, detect deviations, and associate energy consumption with domestic and industrial habits. This will result in poor knowledge about the demand dynamics.
- **Interpretability**: Forecasting models do not have causal relationships, including the effect of weather, calendar effects, or socio-economic variables on demand; they just give point estimates. This constrains the decision-making of stakeholders.
- **Integrated Intelligence**: The current tools offer disjointed information, which only predicts or monitors. It does not have a comprehensive platform with a combination of behavioral analysis, predictive modeling, interpretability, and a decision of support that is necessary as a holistic energy plan.

Technical gaps

1. Lack of integration of various sources of data.

Most forecasting models are based primarily on the past demand of electricity and do not consider other external factors, which are often significant and can be weather, socioeconomic factors, industrial production, changes in policies and supply constraints. These models lack built-in data and thus do not completely capture the complex dynamics of demand and do not react well to unexpected shifts.

2. Weaknesses of existing LSTM-based models.

The LSTM models have been extensively applied in electricity load forecasting due to the characteristics of the sequential nature and the short-term dependencies they can capture. Nonetheless, the vast majority are optimized for an annual and short-term forecast. They have less long-term accuracy that diminishes their use in the planning and policy making of energy.

3. Poor research and analysis of advanced deep learning models.

The other weakness is the restricted scope of model comparisons. Most of the research considers some of the machine learning methods, including MLP and LSTM models. The more complex deep learning architectures, such as GRUs, CNN-LSM hybrids, or attention models based on transformers, have not been explored properly. Consequently, their ability to enhance the capture of the need for patterns of complex electricity demands is still not well studied.

4. Absence of hybrid and ensemble deep learning models.

Hybrid models are not often studied, especially in current research. Architectures can be combined, e.g., CNNs with LSTMs or LSTMs with attention mechanisms, and they can help in making a better forecast by making use of both spatial and temporal correlations of demand data. Despite this prospect, hybrid practices are rarely used.

5. Missing probabilistic forecasting and uncertainty analysis.

Most of the forecasting research evaluates model performance in terms of common measures of accuracy like MAE, MSE, and Root RMSE. Although these metrics are exact, they do not have any data about the uncertainty of the forecast. Probabilistic forecasting and uncertainty quantification are becoming significant to risk management, and operational decision-making processes yet are not often covered in the current models.

[9] [10]

OBJECTIVE

2.1 Main Objective

To create an electricity demand forecasting and decision-support system, tailored to Sri Lanka with the following characteristics: behavioral patterns of consumption, external drivers of demand and presentable, and probabilistic forecasts of energy planning effectiveness

2.2 Specific Objectives

***Behavioral Consumption Analysis*:** Examine and classify common daily and seasonal electrical consumption patterns in the households and industries in Sri Lanka. This involves determining groups of consumption behavior (e.g. weekday vs weekend) and deviations (anomalies) in demand, in order to understand better the underlying causes of changes in load.

***Data Interpretability*:** Add other sources of data (weather variables, calendar effects, economic variables, policy changes) into forecasting models. Determine the cause-effect dynamics of these variables on demand with interpretable ML methodologies (e.g. feature importance or attention scores) so that the stakeholders can identify which changes in demand are attributable to external factors.

***Development of the Advanced and Hybrid Models*:** Develop and compare the advanced deep learning models (GRU, CNN-LSTM hybrids, and Transformer-based) based on the demand data in Sri Lanka. Train hybrid models (e.g. CNN+LSTM, LSTM+Attention) that are more sensitive to spatial/feature correlation and temporal dynamics. Compare the improvements of model accuracy that the LSTM does.

***Probabilistic Forecasting and Uncertainty Quantification*:** Extrapolate optimal models to generate probabilistic forecasts (e.g. based on quantile regression, Monte Carlo dropout, or Bayesian layers) and predict a range on demand. Assess uncertainty in forecasting (through measures such as coverage of prediction intervals) to make use of risk perception in decision making.

METHODOLOGY

Design Science Research (DSR)

The study is based on Design Science Research (DSR). This entails the progressive creation and testing of a product, which in this instance is an adaptive AI model in order to address the issue of non-adaptive, static electricity forecasting. This study is also something experimental,as it compares several hybrid deep learning structures (GRU, CNN-LSTM, Transformers) to identify the most effective and resilient model to apply to the Sri Lankan context

Technical Approaches and Algorithms

To obtain "Adaptive Forecasting," the system will employ the following:

Time-Series Feature Engineering: The system will use CNN layers to obtain spatial/feature correlations and LSTM layers to obtain temporal dynamics.

Attention Mechanisms: Combinations of attention scores used to determine the historical time steps or external factors that have the greatest impact on the forecast.

Adaptation Logic: This will involve automated model adjustment or re-training when changes in patterns (detected in the Behavior Module) are detected.

Probabilistic Forecasting: Applying Quantile Regression or Monte Carlo Dropout to create confidence intervals and estimate uncertainty in demand.

Justified Tools and Platforms

Tools	**Justification**
Python 3.10	Universal deep learning ecosystem — TensorFlow, PyTorch, Pandas, and Scikit-learn all built around Python. No comparable alternative for ML development.
TensorFlow 2.x / Keras	Keras with TensorFlow 2.x was selected for its simplicity, flexibility, and efficient handling of large datasets during model training
PyTorch	Dynamic computation graph makes Monte Carlo Dropout at inference time straightforward — model. Train () kept active during inference to sample dropout masks.
Scikit-learn	Min Max Scaler, Timeseries Split, and MAE/RMSE/MAPE metric functions — standard preprocessing pipeline for ML in Python.
Pandas	Lag feature creation (shift)), rolling statistics (rolling). mean ()), and timestamp parsing all native Pandas operations for the feature engineering pipeline.

NumPy	Constructs sliding window sequences (shape: [samples, timesteps, features]) and aggregates 100 Monte Carlo passes into mean and percentile confidence bounds.
Google Colab Pro	A100/V100 GPU access at USD 9.99/month. Full CNN-LSTM and Transformer training on 189,888 records takes minutes vs hours on local CPU. Already in project budget.
React.js	Component-level state management handles dynamic views — short/long-term forecast toggle, live metric cards, and drift alert banner — cleanly without page reloads.
Tailwind CSS	Utility-first classes produce consistent dashboard spacing and typography without separate stylesheet files.
GitHub	Branch-per-component workflow for all four team members. Pull request merges prevent integration conflicts when components connect at the API layer.
VS Code	Integrated terminal, Python/Pylance extension, and Git support covers both FastAPI backend and React frontend development in one environment.

Table 1:Technologies that will be used

Data Collection and Ethical Considerations

Data Sources: The data sources that will supply the historical data on the daily electricity consumption will be the government sources and energy authorities in Sri Lanka. Data on weather will be obtained through meteorological department records.

Pre-processing: The data will be cleaned to remove inconsistencies and outliers, helping improve the model's accuracy and reliability.

Ethical Implications: This project will use only aggregated and anonymized data, so no personal or sensitive information will be involved. Because of this, formal ethical clearance is not required.

Validation Strategy and Performance Metrics

The model will be validated by use of train-test split and cross-validation which is specifically designed to work with time-series data (e.g., TimeSeriesSplit) to demonstrate its novelty in adapting to changing trends.

- Point Forecast Metrics: MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), MAPE (Mean Absolute Percentage Error) to measure performance.
- Probabilistic Metrics: PICP (Prediction Interval Coverage Probability) to measure the reliability of uncertainty.

HIGH-LEVEL SYSTEM ARCHITECTURE

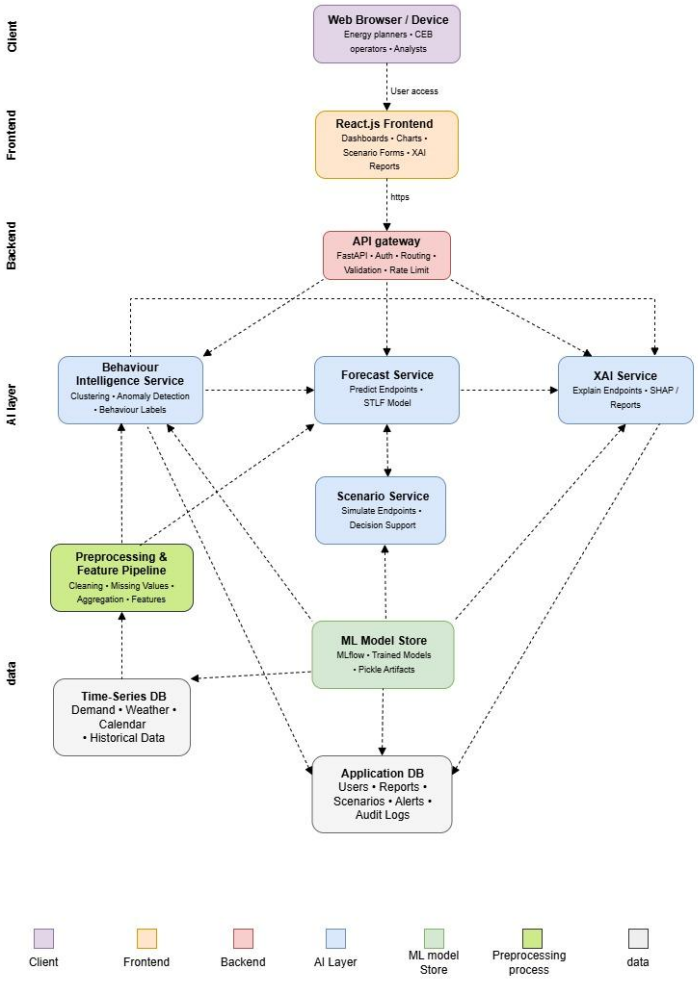


Figure 1: system architecture diagram

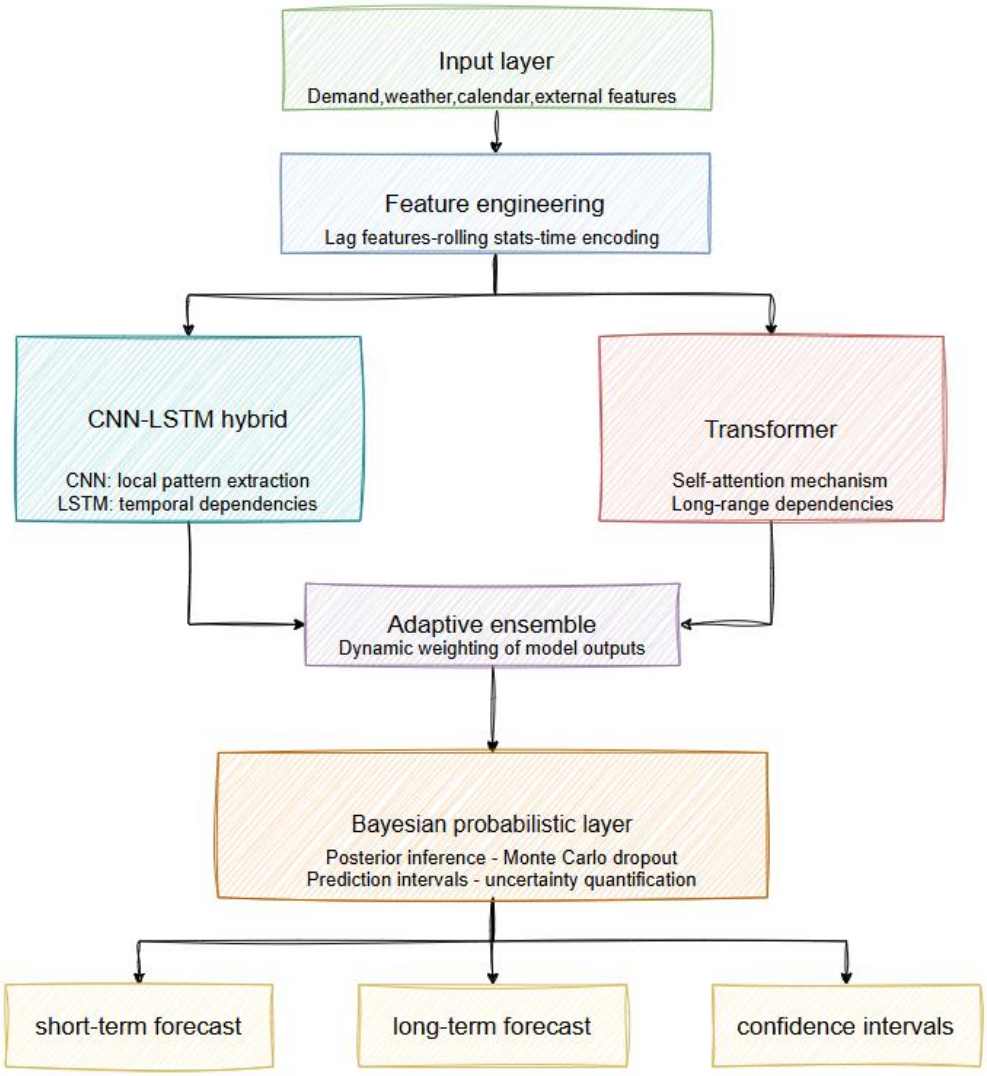


Figure 2:component architecture diagram

USER REQUIREMENTS

Stakeholder Identification

The stakeholders in this system are the individuals who donate information, those who make plans based on the information, and those whose life is impacted by the stability of the grids:

Energy Planners and Analysts: They are the users who should be knowledgeable of the demand behavior and scenario simulation to make long-term strategic decisions. It is upon these that energy planners and analysts base their future plans.

Utility companies, like CEB/LECO in Sri Lanka: They should have the forecast to prevent blackouts in the generation of power and minimize waste. These forecasts are used by utility companies such as CEB/LECO to maintain a stable power grid.

Industrial and Commercial Consumers: When the consumption pattern is analyzed, these entities enjoy an efficient power grid. Industrial and commercial consumers would like to receive information about the impact of their energy consumption on the grid.

Functional Requirements (FR)

These are requirements that specify the functionality of the system:

Data Integration & Pre-Processing: The system has to integrate data providers, such as past electricity demand, weather, calendar effects, and economic factors. This assists the system to be aware of the data.

Pattern Clustering of Behaviors: The system should cluster the consumption patterns of households and industries on a seasonal basis. This can be used to determine energy consumption trends.

Hybrid Predictive Modeling: The system should be capable of giving demand forecasts with the help of a combination of deep learning. This assists in the forecasting of energy demand.

Probabilistic Forecasting: The system should produce a set of predicted demand based on techniques such as Quantile Regression or Bayesian layers. This assists in measuring uncertainty of energy demand.

Non-Functional Requirement (NFR)

Such are the quality attributes and constraints of the systems:

Accuracy: The methods have to be less precise as compared to the forecasting models. Proper predictions assist the consumers in decision making.

Usability (Interpretability): The visualizations and cause effect reports should make the presentation of the AI-driven insights easy to understand. This assists users to get to know the insights.

Scalability: The system should be able to handle quantities of information effectively and deliver prediction in a fair amount of time. This assists the users in getting the insights they require on a short time.

Reliability: The quantification of uncertainty should be sound with the prediction limits embracing the demand. This assists the users to have confidence in the predictions.

Flexibility: The models should be flexible to the shifting trends and behavioral changes, in the Sri Lankan energy market with time. This can assist the system to be accurate and relevant.

COMMERCIALIZATION PLAN

1. Target Market and Customer Persona Target Market: National energy utilities and grid operators in developing countries (in this case, the South Asian region, beginning with Sri Lanka). Secondary Market: Large-scale industrial zones, renewable energy developers, and smart-city infrastructure planners. Customer Persona: The Grid Strategist Role: Operational Engineer or Lead Planner at a utility company (e.g., Ceylon Electricity Board). Pain Points: Can no longer predict load shedding accurately; operations are too expensive, and no one knows why demand spikes occur. Goals: To get 9
2. The system includes an end-to-end intelligence layer, not a simple prediction tool. Moving Black Box to Open Book: Unlike other systems, our system provides adaptive learning, allowing users to see how the weather will affect demand or how a new industrial load will affect it, before it occurs. Proactive Risk Management: Unlike other systems, our system includes adaptive learning, where users can see how the weather will impact demand or how a new industrial load will impact demand before it takes place.
3. Revenue ModelB2G (Business-to-Government) Licensing: Annual subscription-based Software as a Service (SaaS) model to national grid operators. Consultancy & Integration Fees: Revenue one-time to integrate the AI module with the existing SCADA or smart meter infrastructure.
4. Estimation of Costs Development Costs: Cloud computing to run deep-learning models (LSTM, Transformers) and dedicated software libraries. Data Acquisition: Payroll to support server maintenance and constant retraining of models to deal with "concept drift" Operational Costs: Payroll to maintain server availability and keep models retrained.
5. Pricing Strategy Value-Based Pricing: Pricing depends on the savings that are created for the utility. As an illustration, when a 1 percent improvement in accuracy will save USD 1.6M each year, the license will be priced as a percentage of the savings. Pilot Program Pricing: The license will be priced as a percentage of the savings to the local provincial utilities to collect proof of concept and optimize the local behavior models.
6. Competitive Advantage Regional Specialization: Local holidays, tropical weather patterns, and specific economic cycles are all accounted for, which is important because it allows better risk-based planning than standard deterministic models. Uncertainty Quantification: Probabilistic ranges are also provided (not a single number), which is important because it allows risk-based planning to be done.
7. Intellectual Property (IP) Considerations Algorithm Patents: Patent the unique hybrid model structures (i.e. the specific CNN-LSTM or Transformer combinations) designed during the project. Trade Secrets: Copyrighting the source code of the "Explainable AI" reporting engine and the "Scenario Simulation" interface.

BUDGET AND JUSTIFICATION

The high-performance hardware and cloud infrastructure budget means that the platform will be able to scale tensors involving complex time-series without a bottleneck. This is essential to the process of training advanced hybrid models such as CNN-LSTM and Transformers to learn the non-linear demand patterns. Support Data Interpretability Data Interpretability involves dedicated storage and data access, which in turn combines weather, industrial, and calendar variables to generate probabilistic forecasts and quantify uncertainty.

Transportation funds are necessary to verify data anomalies with real-world events, which is done by checking data with utility offices (CEB) and weather stations. This makes the behavioral consumption analysis and scenario simulations realistic and helpful to energy planners. Collectively, these costs can offer the infrastructure required to offer actionable, adaptive intelligence to the energy management of Sri Lanka.

Item	**Quantity / Duration**	**Unit (LKR)**	**Total (LKR)**	**Justification**
Hardware	1	217,000	217,000	Required for local development, feature engineering, model inference testing, and accessing Google Colab Pro remotely.
Cloud computing (Google Colab Pro/GPU	9 Months	4,500	40,500	Training and evaluating ML models using GPU resources
Web hosting/cloud server	9 Months	2500	22,500	Hosting backend APIs and explanation services
Data storage (cloud storage)	9 Months	1500	13,500	Storage for Electricity demand data set, model outputs, and another research-related data.
Domain & SSL certificate	1	5,000	5,000	Secure web access for dashboard and API endpoints
Visualization & Analytics tools	1	6000	6,000	To create interactive dashboards creation and visualizing the reports

Data collection	4 Trips	800	3,200	Travel for dataset collection and Discussions
Miscellaneous	1	4,000	4,000	Research communication, document printing
Total budget			**311,700**	

Table 2: Expected budget and justification

WORK BREAKDOWN STRUCTURE (WBS)

WBS No	Task	Sub task
1.0	Research requirements& literature review	Problem identification Literature review Requirements analysis
2.0	System design	High-level architecture design, Component design, Database & storage design, UI/UX Wireframing
3.0	Dataset preparation	Data acquisition, Data cleaning, Data normalization, Feature engineering
4.0	Model development	CNN-LSTM hybrid model, Transformer model, Bayesian probabilistic layer
5.0	Backend development	API design & setup, Data pipeline integration, Model serving, Security & configuration
6.0	Frontend development	Dashboard layout & routing, Forecast visualization panel
7.0	System integration	Module integration, End-to-end pipeline testing
8.0	Testing & evaluation	Model evaluation, System testing, Usability evaluation
9.0	Documentation & final submission	Code documentation, Dissertation chapter writing, Final report compilation, Final submission

Table 3.: work break down structure

GANTT CHART

System Development Project Timeline – 2026													
		● Semester 1 (Jan – Jun 2026)						● Semester 2 (Jul – Dec 2026)					
Phase	**Activity**	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Phase 1 – Research & Design													
Phase 1	1.0 Research Requirements	·	·										
Phase 1	2.0 System Design		·	·									
Phase 2 – Data & Model													
Phase 2	3.0 Dataset Preparation			·	·	·							
Phase 2	4.0 Model Development					·	·	·					
Phase 3 – Implementation													
Phase 3	5.0 Backend Development							·	·	·			
Phase 3	6.0 Frontend Development									·	·		
Phase 3	7.0 System Integration											·	
Phase 4 – Testing & Delivery													
Phase 4	8.0 Testing & Evaluation												·
Phase 4	9.0 Documentation & Final												·
Summary	**Tasks per Semester**	**Semester 1: Tasks 1.0 – 4.0 | Research → Model Development**						**Semester 2: Tasks 4.0 – 9.0 | Development → Delivery**					

REFERENCE LIST

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[Appendix - A: Summary of Required Dataset]

Summary of Required Dataset from the Resource Personnel:

1. Temporal Data

- Timestamp — date and time of each observation at 15-minute intervals (primary time index)
- Hour of Day — hour of the observation (0 = midnight, 23 = 11 PM)
- Day of Week — numeric day identifier (0 = Monday, 6 = Sunday)
- Month — month of the year (1 = January 12 = December)
- Season — seasonal classification (Summer, Winter, Fall)

2. Target Variable

- Load Demand (kW) — electrical load demand in kilowatts, the primary variable to be forecasted by the CNN-LSTM hybrid and Transformer models

3. Weather & Environmental Data

- Temperature (°C) — average temperature in Celsius during each 15-minute interval
- Humidity (%) — average relative humidity during the forecast period
- Wind Speed (m/s) — average wind speed during the forecast period
- Rainfall (mm) — total rainfall in millimeters during the forecast period
- Solar Irradiance (W/m²) — solar energy received in watts per square meter, capturing the effect of sunlight on electricity demand

4. Socioeconomic & Pricing Data

- GDP (USD) — gross domestic product in USD representing economic activity that influences power demand
- Per Capita Energy Use (kWh) — average energy consumption per person in kilowatt-hours
- Electricity Price (LKR/kWh) — price of electricity in Sri Lankan Rupees per kilowatt-hour during each period

5. Calendar & Event Data

- Public Event — binary flag indicating whether a public event occurred during the interval (1 = Yes, 0 = No)